

Probability and Statistics (MT-2005)

Homework # 9

Properties of Good Estimator

Question # 1

Let $X_1, X_2, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 .

- (a) Determine whether the following estimators are unbiased for μ :

$$\hat{\mu}_1 = \bar{X}, \quad \hat{\mu}_2 = \frac{1}{2}X_1 + \frac{1}{2}X_2, \quad \hat{\mu}_3 = \bar{X} + \frac{1}{n}$$

- (b) Show that the estimator

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is unbiased for σ^2 .

Question # 2

Consider a simple linear regression model of the form:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, 2, \dots, n$$

where ϵ_i are random errors.

- (a) Show that the ordinary least squares (OLS) estimator $\hat{\beta}_1$ of the slope parameter β_1 is unbiased.
(b) Prove that the OLS estimator $\hat{\beta}_0$ of the intercept β_0 is also unbiased.

Hint:

- Recall that the slope estimator is given by:

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

and that $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$.

- When taking expectation, use the linearity of expectation and the fact that $\mathbb{E}[\epsilon_i] = 0$ to show $\mathbb{E}[\hat{\beta}_1] = \beta_1$ and $\mathbb{E}[\hat{\beta}_0] = \beta_0$.
- Assume that $\mathbb{E}[\epsilon_i] = 0$ and that the error terms are independent and identically distributed (i.i.d.) with constant variance.
- Treat X_i and Y_i as fixed constants (non-random) with respect to the expectation operator $\mathbb{E}[\cdot]$.

Question # 3

- (a) Let $X_i \sim \text{Exp}(\lambda)$. Show that $\hat{\lambda}_n = \frac{1}{\bar{X}}$ is a consistent estimator for λ .
- (b) Let X_1 be a random sample from a population with mean μ and variance σ^2 . Is the estimator $\hat{\mu} = X_1$ consistent for the population mean μ ? Justify.

Question # 4

- (a) Let $\hat{\theta}_1$ and $\hat{\theta}_2$ be two unbiased estimators of θ with variances $\frac{4}{n}$ and $\frac{5}{n}$, respectively. Which is more efficient? Find the relative efficiency, where The relative efficiency of two estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ for a parameter θ is given by:

$$RE(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

- (b) For a sample from $N(\mu, \sigma^2)$, compare the efficiency of:

$$\hat{\mu}_1 = \bar{X}, \quad \hat{\mu}_2 = \frac{1}{3}(X_1 + X_2 + X_3)$$